

CH 1 — Introductio to Macroeconomics

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1.1

Definition — Recession

A recession is a broad decline in economic activity. Because households and firms are connected through spending and income, a disruption in one part of the economy can spread and become an aggregate decline.

- The **direct effect** falls on the households and firms initially affected by the disruption
- The **indirect effect** develops as their losses change the spending and income of others.

Definition — Basic Terms

Exogenous: outside or determined outside of the model

Endogenous: determined within the model

Faillacy of composition is the error of assuming that a conclusion that is true for one part must also be true for the system as a whole.

Across the economy, household spending becomes business revenue, while firms pay wages and other income to households

- Consumers sells their labour service to the firms and purchase goods.
- Firms produces a single good and pays wages to the consumers.

Macroeconomic outcomes result from individual decisions and the ways those decisions interact.

Example:

Paradox of thrift(fallacy of composition)

If individual savings increase, it may not mean that total aggregate savings would incread.

If each consumer increase their savings rate will aggregate savings increase?

Maybe or maybe not.

Consider an economy with no government and no trade. All output is either consmed by households or bought by firms as investment, so $Y = C + I$.

Y = production, C = consumption, S = savings, I = investment

$$Y = C + S = C + I \implies S = I \implies Y = C + I$$

	Households save 10%	Households save 20%
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Consumption rule	$C = 0.9Y$	$C = 0.8Y$
Income Equation	$Y = 0.9Y + 100$	$Y = 0.8Y + 100$
Income	1000	500
Consumption	900	400
Saving	100	100

Here saving rate doubled, aggregate saving does not move, but income falls.

But the paradox of thrift may not always apply. Saving rate and aggregated saving rate both went up as disposable income becomes forced savings and precautionary savings.

1.2 Data Statistics

Frequencies: the rate at which data is observed:

- High frequency: stock market, bond market, interest rates, exchange rate
- Medium frequency: CPI, unemployment, GDP(quarterly)
- Low frequency: Census, Annual releases

Definition – Measures of the Economy

1. **Real GDP:** total production adjusted for price changes
2. **Unemployment rate:** the share of the labour force without work and actively seeking it
3. **CPI inflation:** change in the cost of a representative consumer basket
4. **Policy interest rate:** the Bank of Canada's target overnight rate
5. **Real GDP per capita:** real output per person

Indicators in Business Cycle

Procyclical variable moves in the same direction as real GDP. Employment is usually procyclical.

Countercyclical variable moves the opposite direction as real GDP. Unemployment is usually countercyclical.

Acyclical variable is not affected by real GDP

Definition – Recession

Expansion: output is rising

Peak: the expansion ends and activity to fall

Recession: activity declines from the peak to the trough

Through: the recession ends and activity begins to rise

Recovery : the early expansion before output regains its previous peak